

```
create database suubddb;
```

```
use suubddb;
```

```
CREATE TABLE departments (  
dept_id INT PRIMARY KEY,  
dept_name VARCHAR(50),  
location VARCHAR(50)  
);
```

```
INSERT INTO departments VALUES  
(10,'HR','Jaipur'),  
(20,'IT','Bangalore'),  
(30,'Finance','Mumbai'),  
(40,'Marketing','Delhi'),  
(50,'Legal','Pune');
```

```
CREATE TABLE employees (  
emp_id INT PRIMARY KEY,  
emp_name VARCHAR(50),  
dept_id INT,  
salary INT,  
city VARCHAR(50)  
);
```

```
INSERT INTO employees VALUES  
(101,'Amit',10,45000,'Jaipur'),  
(102,'Bhavna',20,72000,'Bangalore'),  
(103,'Chirag',20,68000,'Bangalore'),  
(104,'Divya',30,55000,'Mumbai'),  
(105,'Esha',NULL,39000,'Delhi'),  
(106,'Farhan',40,61000,'Delhi'),
```

```
(107,'Gaurav',60,50000,'Kolkata'),
```

```
(108,'Hina',30,83000,'Mumbai');
```

```
CREATE TABLE projects (
```

```
proj_id INT PRIMARY KEY,
```

```
proj_name VARCHAR(50),
```

```
dept_id INT,
```

```
budget INT
```

```
);
```

```
INSERT INTO projects VALUES
```

```
(1,'Payroll Revamp',10,120000),
```

```
(2,'Cloud Migration',20,500000),
```

```
(3,'Mobile App',20,300000),
```

```
(4,'Audit Automation',30,250000),
```

```
(5,'Brand Refresh',40,180000),
```

```
(6,'Data Lake',70,400000);
```

```
select e.emp_id, e.emp_name, d.dept_name, d.dept_location from employees as e join
```

```
departments as d on e.dept_id=d.dept_id; -- 1
```

```
SELECT p.ProjectName, d.DepartmentName
```

```
FROM Projects as p
```

```
INNER JOIN Departments as d
```

```
ON p.DepartmentID = d.DepartmentID; -- 2
```

```
SELECT e.EmployeeName, p.ProjectName
```

```
FROM Employees as e
```

join department as d

JOIN Projects as p

ON e.DepartmentID = p.DepartmentID; -- 3

SELECT e.EmployeeName, d.DepartmentName

FROM Employees e

LEFT JOIN Departments d

ON e.DepartmentID = d.DepartmentID; -- 4

SELECT p.ProjectName, d.DepartmentName

FROM Projects as p

left JOIN Departments as d

ON p.DepartmentID = d.DepartmentID; -- 5

SELECT e.EmployeeName, d.DepartmentName

FROM Employees e

left JOIN Departments d

ON e.DepartmentID = d.DepartmentID

where d.department_name is null; -- 6

SELECT e.EmployeeName, d.DepartmentName

FROM Employees e

RIGHT JOIN Departments d

ON e.DepartmentID = d.DepartmentID; -- 7

```
SELECT p.ProjectName, d.DepartmentName
FROM Projects p
LEFT JOIN Departments d
ON p.DepartmentID = d.DepartmentID; -- 8
```

```
SELECT d.DepartmentName
FROM Departments d
LEFT JOIN Employees e
ON d.DepartmentID = e.DepartmentID
WHERE e.EmployeeID IS NULL; -- 9
```

```
select * from employees as a cross join department as a where (dept_name='legal' or
dept_name='finace') and salary>70000; -- 10
```

```
select * from proects cross join department
where budget>=40000 and location in ('mumbai','pune'); -- 11
```

```
SELECT COUNT(*) AS Total_Combinations
FROM Employees
CROSS JOIN Departments; -- 12
```

```
SELECT EmployeeName, Salary
FROM Employees
WHERE Salary > 60000; -- 13
```

```
SELECT *
FROM Employees
```

WHERE City = 'Delhi'

OR City = 'Mumbai'; -- 14

SELECT *

FROM Employees

WHERE Dept_ID IS NULL; -- 15

SELECT *

FROM Departments

WHERE Location IN ('Bangalore', 'Delhi'); -- 16

SELECT *

FROM Departments

WHERE Dept_ID >= 30; -- 17

SELECT *

FROM departments

WHERE dept_name LIKE 'F%'

OR dept_name LIKE 'L%'; -- 18

SELECT *

FROM projects

WHERE budget BETWEEN 200000 AND 400000; -- 19

SELECT *

FROM projects

WHERE dept_id = 20; -- 20

```
SELECT *  
  
FROM projects  
  
WHERE project_name LIKE '%a%'  
  
AND budget < 300000; -- 21
```

```
SELECT  
  
    e.emp_name,  
  
    d.dept_name,  
  
    d.location,  
  
    p.project_name  
FROM employees e  
INNER JOIN departments d  
    ON e.dept_id = d.dept_id  
INNER JOIN projects p  
    ON d.dept_id = p.dept_id; -- 22
```

```
SELECT  
  
    e.emp_name,  
  
    d.dept_name,  
  
    d.location,  
  
    p.project_name  
FROM employees e  
LEFT JOIN departments d  
    ON e.dept_id = d.dept_id  
LEFT JOIN projects p  
    ON d.dept_id = p.dept_id; -- 23
```

```
SELECT
    e.emp_name,
    e.salary,
    d.dept_name,
    p.project_name,
    p.budget
FROM employees e
INNER JOIN departments d
    ON e.dept_id = d.dept_id
INNER JOIN projects p
    ON d.dept_id = p.dept_id
WHERE e.salary > 60000
AND p.budget > 250000; -- 24
```

```
SELECT
    d.dept_name,
    d.location,
    e.emp_name,
    p.project_name
FROM departments d
LEFT JOIN employees e
    ON d.dept_id = e.dept_id
LEFT JOIN projects p
    ON d.dept_id = p.dept_id; -- 25
```

```
SELECT
    e.emp_name,
```

```
d.dept_name,  
d.location,  
p.project_name  
FROM employees e  
INNER JOIN departments d  
    ON e.dept_id = d.dept_id  
INNER JOIN projects p  
    ON d.dept_id = p.dept_id  
WHERE d.location IN ('Bangalore', 'Mumbai'); -- 26
```

```
SELECT  
    e1.emp_name AS Employee_1,  
    e2.emp_name AS Employee_2,  
    e1.city  
FROM employees e1  
INNER JOIN employees e2  
    ON e1.city = e2.city  
AND e1.emp_id < e2.emp_id; -- 27
```

```
SELECT  
    p.project_name,  
    d.dept_name,  
    e.emp_name  
FROM projects p  
LEFT JOIN departments d  
    ON p.dept_id = d.dept_id
```

LEFT JOIN employees e

ON d.dept_id = e.dept_id; -- 28